Dharmik Umeshkumar Jha

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EDUCATION

Nirma University, Institute of Technology, Ahmedabad, Gujarat, India. *Expected Graduation: 2027*

- **B.Tech Mechanical Engineering** (Minor: Data Science) — **CGPA: 8.64/10.0**
- **Relevant Coursework:** Machine Design, Finite Element Analysis, Design & Dynamics of Machines, Aerodynamics, Automation & Control, Manufacturing Processes

EXPERIENCE

Mechanical Design Intern — Shree Siddhi Vinayak Engg. Co. *May 2026 - July 2026*
Industrial Internship

- Developed concept designs for an agricultural rover and compared rigid, rocker and independent-arm mechanism options against the given size constraints.
- Evaluated each option on layout feasibility and manufacturability to support the choice of concept.

Society of Automotive Engineers (SAE) Nirma Collegiate Club — Team Arrow *Oct 2023 – Mar 2026*
Captain (DDC 2025, Aero East 2026) | Design Lead (Aerothon 2025) | Junior Designer (DDC 2024)

- **1st of 100+ teams, SAEISS DDC 2025:** led a team of 8 on a ∼10 kg MTOW UAV; cut structural weight ∼12% by topology optimisation; FEA-validated landing gear and wing.
- **11th of 75 international teams, SAE Aero Design West 2025:** member of aerodynamics and structures for the micro-class UAV; ∼20% rib mass saving.
- **Aerothon 2025 (Design Lead):** generative-designed a single-piece 385 g Nylon-12 SLS quadrotor frame; the final build hovered for 21 min.
- **Aero East 2026:** led the team on a tilt-rotor VTOL tricopter; built an MDF prototype to test the tilt mechanism before integrating it into the UAV.

Volunteer, Organising Support: ICMPC 2024 and FMFP 2025 *June 2024 & Dec 2025*
Nirma University, Ahmedabad

- Supported conference activities at the 16th International Conference on Material Processing and Characterisation (ICMPC 2024) and the 12th International & 52nd National Conference on Fluid Mechanics and Fluid Power (FMFP 2025), both hosted at Nirma University.

RESEARCH PUBLICATION

- **Journal Paper:** *"Design and Aerodynamic Performance Assessment of High Payload Capacity Fixed-Wing Unmanned Aerial Vehicle (UAV)."*
  Published in the *Journal of Flow Visualization and Image Processing*, ***Begell House***.
  DOI: 10.1615/JFlowVisImageProc.2026064024
- **Conference Paper:** *"CFD Integrated Design of Unmanned Aerial Vehicle for Higher Payloads."*
  Presented at *12th International & 52nd National Conference on Fluid Mechanics and Fluid Power (FMFP 2025)*, Nirma University.

SIGNIFICANT ACADEMIC PROJECTS

Full write-ups & additional projects: portfolio

Two-Stage Helical Gear Reduction Gearbox *March 2026 – April 2026*

- Designed a two-stage helical gearbox, sizing gears using Lewis bending and wear strength criteria.
- Designed shafts under combined loading using fatigue analysis and DE-Goodman criterion; selected bearings based on load capacity and L10 life.
- Developed the complete SolidWorks assembly and manufacturing drawings with fits, tolerances, and DFM considerations.

Design and Development of Advanced Class UAV – SAE Aero Design East 2026 *Sept 2025 – March 2026*

- Designed a tilt-rotor tricopter (800 mm wingspan, 1.6 kg AUW) for autonomous hover-to-cruise transition, with a mechanically actuated thrust-vectoring system that rotates the front thrust vectors 110°.

Aerothon 2025 – Disaster Response Quadrotor UAV *March 2025 – August 2025*

- Used three-phase Fusion 360 generative design to produce a single-piece Nylon-12 SLS airframe on a 440 mm wheelbase, optimised for strength-to-weight ratio and manufacturability.
- Ran multiple studies including a 15 m free-fall; final build: 1.85 kg AUW, 3.86 thrust-to-weight, 21 min hover endurance.

Design of Micro Class Aircraft – SAE Aero Design West 2025 *Sept 2024 – April 2025*

- Designed, fabricated, and tested a 1.6 kg micro-class UAV achieving a payload fraction >0.5 under a 450 W power limit. Achieved short takeoff (<10 ft), 5-minute endurance, and high structural integrity through aerodynamic optimisation and validation testing.

Design of Regular Class Fixed Wing Aircraft – SAE Drone Development Challenge *Oct 2023 – March 2025*

- Designed, fabricated, and tested a fixed-wing UAV with empty weight <4 kg, optimised for high payload capacity within geometric constraints (wingspan <72 in; total dimensions <150 in). Constructed using balsa and aeroply with aluminium and acrylic spars; payload box with 500 g airdrop in a conventional tail configuration.

Design of Primary Landing Gear – SAE Drone Development Challenge *Oct 2023 – March 2025*

- Designed a lightweight primary landing gear for a 10 kg fixed-wing UAV using carbon-fibre rods secured with CNC-machined aluminium clamps. Integrated an expanding spring mechanism for landing impact absorption. Final assembly weighed $\sim$250 g - representing less than 2.5% of total aircraft weight- while sustaining rated landing impact loads validated through drop testing.

TECHNICAL SKILLS

- **Mechanical Design & CAD:** SolidWorks, Fusion 360, Generative Design, CAD Modelling, Assembly Design
- **Structural Analysis:** ANSYS Structural, Structural Sizing, Finite Element Analysis
- **Manufacturing & Prototyping:** 3D Printing, Laser Cutting, Composite Fabrication
- **Programming & Numerical Analysis:** Python (NumPy, Matplotlib), MATLAB
- **Flight Systems & Integration:** Pixhawk Autopilot, Mission Planner

HONOURS & AWARDS

- Achieved **overall 11th place** and **5th place in mission score** out of 75 international teams at SAE Aero Design West (Micro Class) 2025, Apollo XI RC Field, Van Nuys, California, USA — April 2025.
- Secured **1st position** in the Regular Class Category and **3rd position** in the Micro Class Category at the SAEISS Drone Design Challenge 2025, Karpaga Vinayaga College of Engineering & Technology, Chennai — March 2025.
- Achieved **overall 4th place** and **2nd Best Design Report** in the Regular Class Category at the SAEISS Drone Design Challenge 2024, SRM University, Chennai — September 2024.